

A Project Report on

CREDIT RISK PREDICTION FOR RETAIL BANKING

- Using the UCI Statlog German Credit Dataset

Submitted in partial fulfillment of the requirements for
Bachelor of Computer Applications (Data Analytics)
MACHINE LEARNING LAB
Paper Code: BCADA 5P 126

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CERTIFICATE

This is to certify that the project report entitled “**Credit Risk Prediction for Retail Banking Using the UCI German Credit Dataset**” is a bonafide record of the work carried out by the students of Group 9 as part of the Machine Learning Lab, Paper Code BCADA 5P126, prescribed for the Bachelor of Computer Applications (Data Analytics) programme under the Department of Advanced Computing, St. Joseph’s University, Bengaluru, during the academic year 2026–2027.

The project has been completed under the guidance and supervision of the concerned faculty members.

Teacher In-charge

Head of the Department

DECLARATION

The members of Group 9 hereby declare that the project report entitled “**Credit Risk Prediction for Retail Banking Using the UCI German Credit Dataset**” is an original academic work completed as part of the Machine Learning Lab course, BCADA 5P126.

The work presented in this report has not been submitted, either wholly or partially, for the award of any other degree, diploma, or academic certification. All datasets, software libraries, technical references, external resources, and supporting materials used in the project have been appropriately acknowledged.

Place: Bengaluru

Date: September 29, 2026

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ACKNOWLEDGEMENT

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The group acknowledges the guidance provided by the faculty members and project coordinators throughout the project lifecycle. Their feedback supported the development of a structured, reproducible, and business-oriented machine learning solution.

The group also acknowledges the UCI Machine Learning Repository for making the Statlog German Credit Data publicly available for academic research and learning. Finally, appreciation is extended to all group members for their individual contributions in data preparation, exploratory analysis, machine learning, inference engineering, business analytics, and dashboard development.

ABSTRACT

This project develops a machine learning system for predicting credit risk in retail banking using the UCI Statlog German Credit dataset. The dataset contains 1,000 records with 20 input variables representing applicants' financial and personal information. Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting models were evaluated, with Gradient Boosting achieving the best performance. A cost-sensitive threshold of 0.13 was used to classify applicants as suitable for approval, manual review, or rejection. The project combines data preprocessing, exploratory analysis, model evaluation, risk-based decision-making, and Power BI visualization to support effective lending decisions.

Keywords: Credit Risk Prediction, German Credit Dataset, Gradient Boosting, Machine Learning, Cost-Sensitive Classification, Retail Banking.

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CHAPTER 01 INTRODUCTION

1.1 Overview of the Project

Credit risk prediction is an important application of machine learning in the financial-services industry. Banks and lending institutions must decide whether an applicant is likely to repay a loan or is likely to default. Such decisions affect profitability, customer satisfaction, regulatory compliance, and long-term portfolio quality.

The present project was developed for a simulated retail banking client named CrediTrust Capital. The project uses historical applicant information to predict whether an individual belongs to the Good Credit Risk or Bad Credit Risk category. The solution was designed as a consultancy-style machine learning project involving data engineering, analysis, modelling, inference, analytics, and dashboard visualization.

The completed pipeline transforms raw applicant information into a risk prediction and then into an operational lending decision. The possible decisions are Approve, Manual Review, and Reject.

1.2 Problem Statement

Banks face financial risk when they approve loans for customers who later fail to repay. At the same time, rejecting reliable customers may reduce revenue, customer acquisition, and future business opportunities.

The project aims to predict the credit risk of loan applicants using 20 financial and personal attributes. The target variable is Credit Risk, where applicants are classified as either Good or Bad credit risk.

The project assumes that approving a customer who later defaults has a financial cost five times greater than rejecting a customer who would have repaid successfully. Therefore, the project treats credit-risk prediction as both a classification problem and a cost-sensitive decision problem.

1.3 Objectives of the Project

- To acquire and understand the UCI Statlog German Credit dataset.
- To validate, clean, and preprocess credit applicant data.
- To identify class imbalance, outliers, and feature relationships.

- To prevent data leakage during preprocessing and model training.
- To train and compare multiple classification algorithms.
- To select a final model using accuracy, precision, recall, and F1-score.
- To develop a cost-sensitive decision threshold for loan approval.
- To create an inference pipeline for scoring new applicants.
- To build a dashboard for visualizing credit-risk KPIs.
- To identify project limitations and future improvements.

1.4 Scope of the Project

The project covers dataset preparation, exploratory analysis, preprocessing, feature transformation, model training, cross-validation, hyperparameter tuning, evaluation, threshold tuning, inference workflow, and dashboard development.

The project does not include live banking integration, production database connectivity, credit-bureau integration, or real-time deployment. The assumed cost values are used for academic analysis and should be validated with actual institutional data before production use.

1.5 End User, Stakeholder, and Project Value

The intended stakeholder is the retail credit team of CrediTrust Capital.

- Loan officers can evaluate applicant-level credit risk.
- Credit managers can monitor portfolio risk and approval trends.
- Risk analysts can identify high-risk customer segments.
- Operations managers can route borderline cases for manual review.
- Senior management can view business KPIs through a dashboard.

The project connects machine learning predictions with financial decision-making. It uses the predicted probability of default and the cost of incorrect decisions to produce a business-oriented lending recommendation.

CHAPTER 2

DATASET AND DATA UNDERSTANDING

2.1 Dataset Description:

The UCI Statlog German Credit Dataset contains 1,000 loan-applicant records, 20 input variables, and one target variable, **Credit_Risk**. The features include financial, personal, employment, housing, and credit-history information.

The target classifies applicants as **Good Credit Risk** or **Bad Credit Risk**. Approximately 70% of applicants are classified as Good Risk and 30% as Bad Risk. Categorical variables were converted using One-Hot Encoding, while numerical variables were standardized using StandardScaler.

The dataset was divided into 80% training data and 20% testing data using a stratified split. Although suitable for academic analysis, the dataset is limited because it is small and does not include modern banking information such as credit scores, transaction history, or actual repayment losses.

Table 2.1: Dataset Summary

Dataset Property	Description
Dataset Name	Statlog German Credit Data
Domain	Finance and Retail
Banking Number of Records	1,000
Number of Input Features	20
Number of Original Columns	21
Target Variable	Credit-Risk
Machine Learning Problem	Binary Classification
Train-Test Split	80:20 Stratified Split
Final Selected Model	Gradient Boosting

2.2 Dataset Source

The dataset was obtained from the UCI Machine Learning Repository.

Table 2.2: Dataset Source Information

Item	Details
Dataset Repository	UCI Machine Learning Repository
Dataset Name	Statlog German Credit Data
Domain	Finance and Retail Banking
Original Contributor	Hans Hofmann
Number of Records	1,000
Number of Input Features	20
Target Variable	Credit_Risk
Problem Type	Binary Classification
Dataset URL	https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data
Purpose of Use	Academic Machine Learning Lab Project

2.3 Features, Variables, and Target

Table 2.3: Feature Groups

Category	Variables
Account Information	Checking Status, Savings Status
Loan Details	Duration Months, Credit Amount, Purpose, Installment
Rate Credit History	Credit History, Existing Credits, Other Installment Plans
Employment Information	Employment Since, Job
Personal Information	Age, Personal Status and Sex, Dependents, Foreign
Worker Housing and Assets	Housing, Property, Residence Since
Other Variables	Telephone, Other Debtors or Guarantors

The target variable is Credit Risk.

Table 2.4: Target Variable Encoding

Target Code	Interpretation
1	Good Credit Risk
2	Bad Credit Risk

2.4 Class Distribution and Data Distribution

The dataset contains approximately 700 Good credit-risk applicants and 300 Bad credit-risk applicants.

Table 2.5: Credit-Risk Class Distribution

Credit Risk Category	Count	Percentage
Good Credit Risk	700	70%
Bad Credit Risk	300	30%

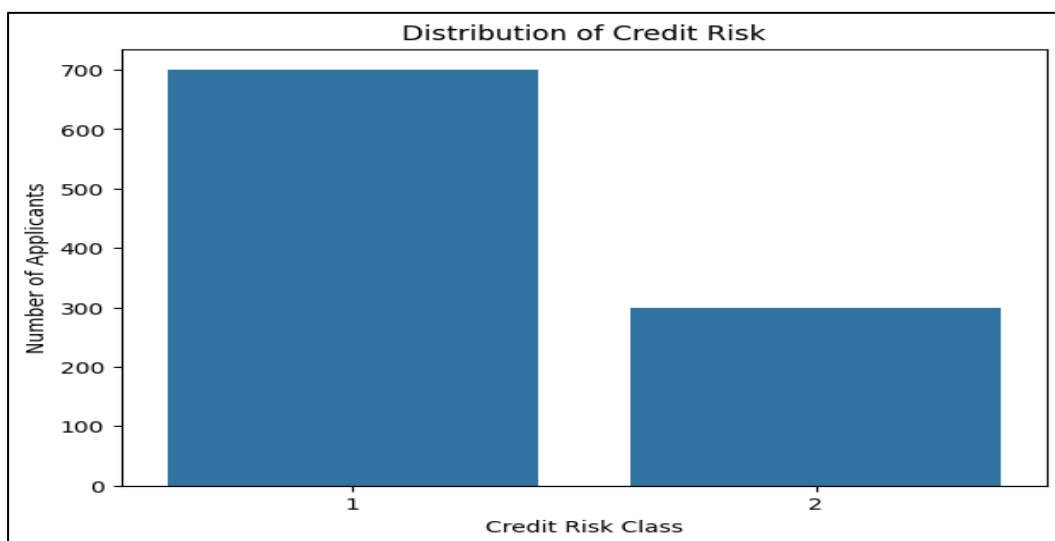


Figure 2.1: Distribution of Good and Bad credit-risk applicants.

The class distribution indicates moderate imbalance. Accuracy alone is therefore insufficient because a model predicting every applicant as Good would obtain approximately 70% accuracy while failing to detect risky applicants.

2.5 Dataset Limitations

- The dataset contains only 1,000 records.
- It represents historical data and may not reflect current banking behaviour.
- It may not represent Indian banking customers.
- It does not include current credit scores or transaction history.
- It does not contain actual loan-recovery or loss-given-default values.
- The dataset requires additional fairness and bias analysis before production use.

CHAPTER 3

DATA PREPARATION AND EXPLORATORY DATA ANALYSIS

3.1 Data Cleaning

The following data-cleaning activities were performed:

- Dataset structure and dimensions were inspected.
- Column names and data types were verified.
- Missing values were checked.
- Duplicate records were checked.
- Required schema columns were validated.
- Invalid values in age, duration, and credit amount were examined.
- Potential outliers were identified using the IQR method.

The detected outliers were not automatically removed because they may represent valid applicants with high loan amounts, long repayment durations, or unusual personal characteristic

3.2 Preprocessing

Table 3.1: Preprocessing Techniques

Feature Type	Technique	Purpose
Categorical Variables	One-Hot Encoding	Converts category values into numerical indicator columns.
Numerical Variables	StandardScaler	Standardizes numerical features to a common scale.
Target Variable	Separated before transformation	Prevents target leakage during preprocessing.
Train-Test Data	Stratified Split	Preserves the Good and Bad class distribution.

Table 3.2: Training and Testing Dataset Sizes

Dataset Portion	Records	Purpose
Training Set	800	Model training and cross-validation
Testing Set	200	Final unbiased evaluation

After One-Hot Encoding, the 20 original variables expanded to 61 machine-learning-ready features.

3.3 Feature Engineering

The primary feature-engineering techniques were:

- One-Hot Encoding of categorical variables.
- Standardization of numerical variables.
- Separation of the target before transformation.
- Preservation of the original variables.
- Generation of 61 processed features from 20 original features.

3.4 Exploratory Data Analysis

3.4.1 Numerical Feature Skewness

Table 3.3: Skewness of Numerical Features

Feature	Skewness
Dependents	1.858
Credit Amount	1.826
Existing Credits	1.192
Duration Months	1.047
Age	1.043
Residence Since	-0.264
Installment Rate	-0.488

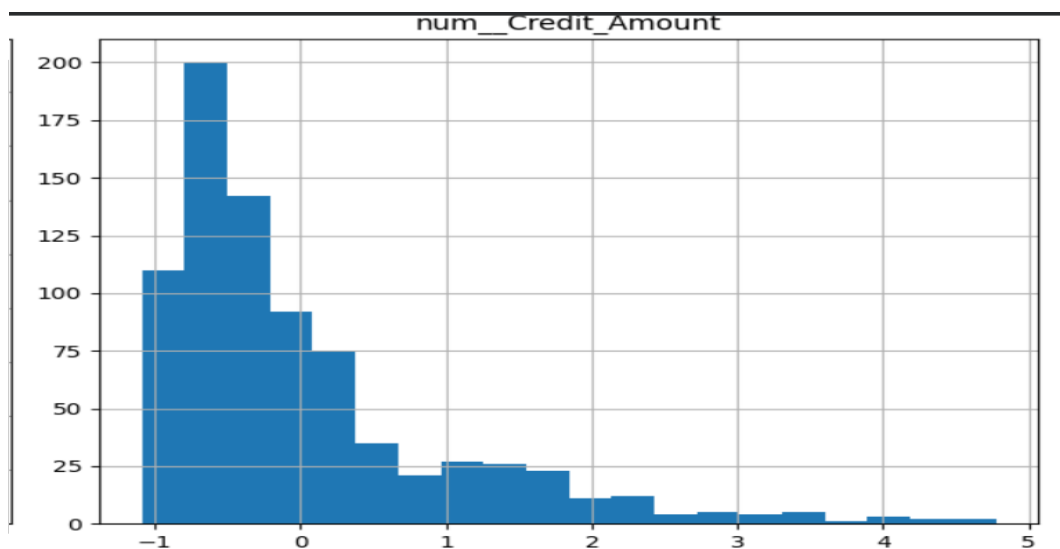


Figure 3.1: Distribution of credit amount among loan applicants.

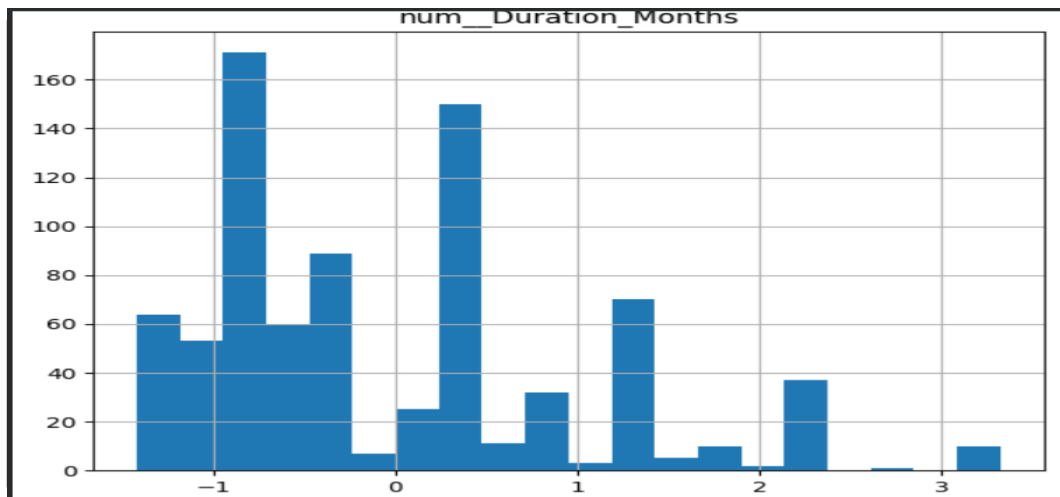


Figure 3.2: Distribution of loan duration in months

Credit amount and loan duration showed right-skewed distributions. This means that most applicants had low or moderate values, while a smaller number had much larger values

3.4.2 Outlier Detection

$$IQR = Q_3 - Q_1$$

$$\text{Lower Limit} = Q_1 - 1.5 \times IQR$$

$$\text{Upper Limit} = Q_3 + 1.5 \times IQR$$

Table 3.4: Potential Outliers Identified

Feature	Potential Outliers
Credit Amount	72
Duration Months	70
Age	23

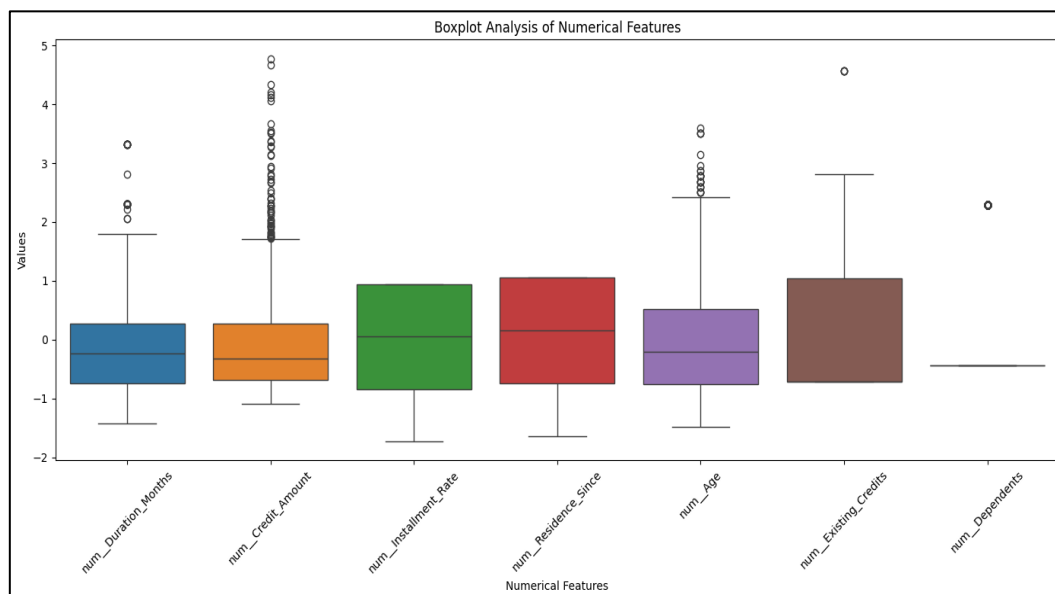


Figure 3.3: Box plots showing potential outliers in credit amount, loan duration, and age.

3.4.3 Correlation Analysis

Table 3.5: Important Numerical Correlations

Feature Pair	Correlation	Interpretation
Duration Months and Credit Amount	+0.638	Longer loans tend to have higher credit amounts.
Credit Amount and Installment Rate	-0.241	Moderate negative relationship.
Residence Since and Age	+0.238	Weak positive relationship.
Age and Existing Credits	+0.148	Weak positive relationship.

Figure 3.4: Correlation heatmap of numerical variables

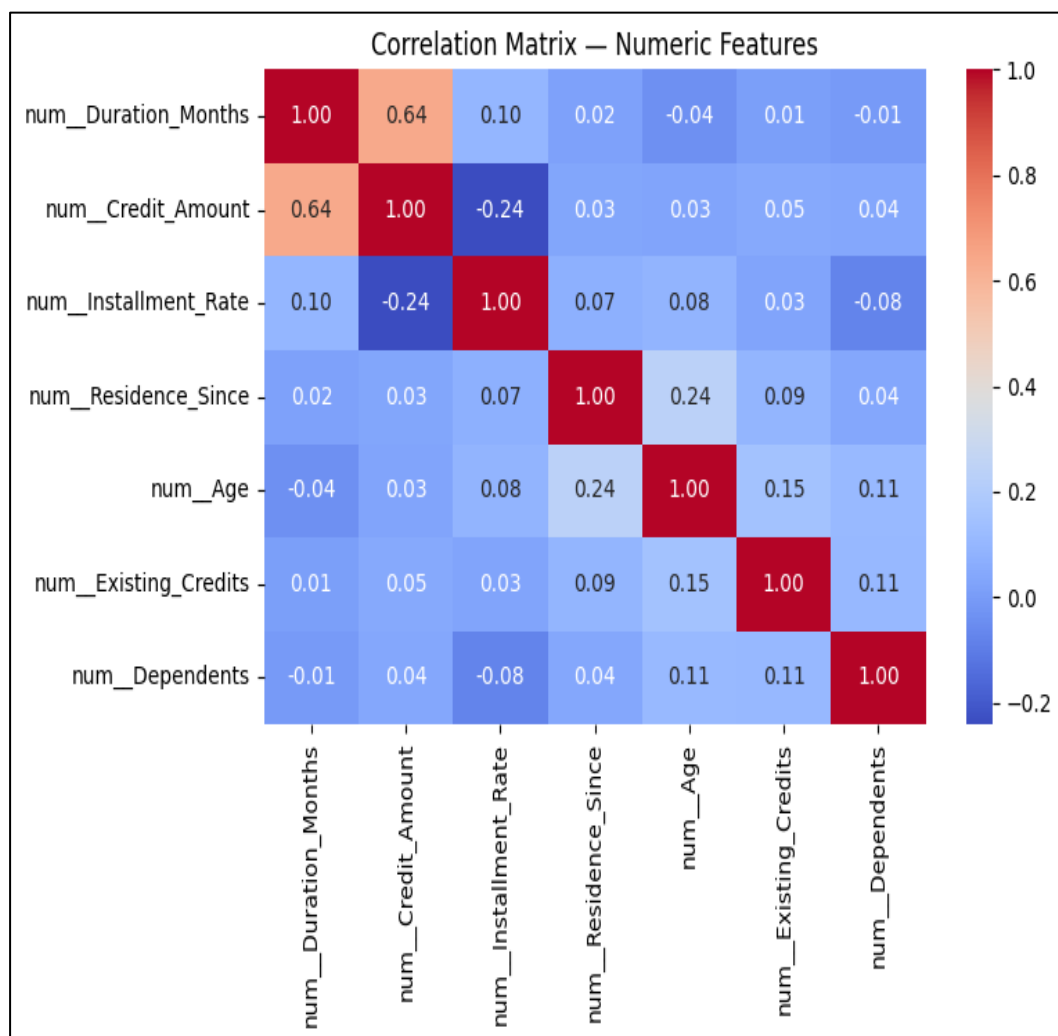
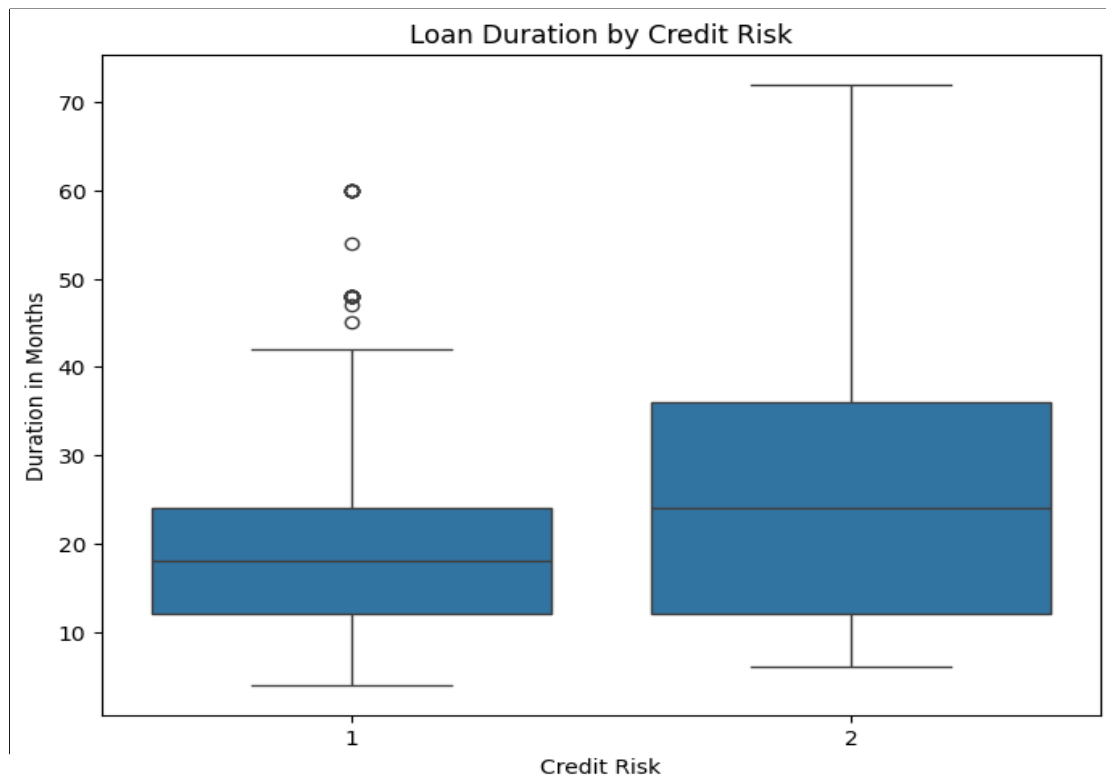


Figure 3.4: Correlation heatmap of numerical variables.



3.5 Key Insights from EDA

- The dataset has a 70:30 Good-to-Bad credit-risk distribution.
- Checking account status showed useful variation across risk classes.
- Credit amount and loan duration had a positive correlation of 0.638.
- Credit amount, duration, and age contained potential outliers.
- The dataset required separate processing for categorical and numerical variables.
- The final preprocessing stage produced 61 numerical features.

CHAPTER 4

MACHINE LEARNING METHODOLOGY

4.1 Machine Learning Problem Type

The project is a binary classification problem. The model predicts whether an applicant belongs to the Good Credit Risk class or the Bad Credit Risk class.

The model also generates a probability score for the Bad Credit Risk class. This probability is used by the decision layer to recommend approval, manual review, or rejection.

4.2 Train-Test and Validation Strategy

The dataset was split into training and testing data using an 80:20 stratified split.

1. The target variable was separated from the input variables.
2. The dataset was divided into 800 training records and 200 testing records.
3. The preprocessing transformer was fitted only on the training data.
4. Five-fold stratified cross-validation was used during model tuning.
5. The testing data was used only for final evaluation.
6. The threshold was tuned using training-set out-of-fold predictions.

4.3 Algorithms and Models Used

Table 4.1: Classification Algorithms

Model	Description	Purpose
Logistic Regression classification	A linear algorithm.	Baseline and interpretable model.
Decision Tree	A rule-based tree classifier.	Captures non-linear patterns.
Random Forest	An ensemble of decision trees.	Reduces overfitting and improves robustness.
Gradient Boosting based	A sequential tree-ensemble.	Captures complex patterns in tabular data.

4.4 Model Training

All models were trained using the same processed training dataset. The test dataset was not used during training or preprocessing.

The model-training process involved:

- Preparing the encoded and scaled training matrix.
- Fitting each candidate model.
- Generating predictions for the test dataset.
- Calculating evaluation metrics.
- Selecting the best-performing model.

4.5 Hyperparameter Tuning

Gradient Boosting was selected for hyperparameter tuning because it produced the strongest initial results. GridSearchCV with five-fold stratified cross-validation was used to identify a suitable combination of hyperparameters.

4.6 Mathematical and Statistical Concepts

4.6.1 Accuracy

Accuracy measures the proportion of correctly classified instances out of the total number of instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

4.6.2 Precision

Precision measures the proportion of correctly predicted positive instances among all instances predicted as positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

4.6.3 Recall

Recall measures the proportion of actual positive instances that were correctly identified by the model

$$\text{Recall} = \frac{TP}{TP + FN}$$

4.6.4 F1-Score:

F1-score is the harmonic mean of Precision and Recall.

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

4.6.5 Cost-Sensitive Threshold

$$\text{Cost-Optimal Threshold} = \frac{C_{FP}}{C_{FP} + C_{FN}}$$

incorrect predictions.

Where:

- C_{FP} = Cost of a false positive.
- C_{FN} = Cost of a false negative.

CHAPTER 5

MODEL EVALUATION AND RESULTS

5.1 Evaluation Metrics

The models were evaluated using Accuracy, Precision, Recall, and F1-score. A confusion matrix and cost analysis were also used to examine the practical impact of the predictions.

5.2 Model Comparison

Table 5.1: Comparison of Classification Models

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	78.00%	74.12%	70.95%	72.10%
Decision Tree	63.00%	55.59%	55.48%	55.52%
Random Forest	74.00%	68.84%	63.81%	64.86%
Gradient Boosting	79.00%	75.33%	72.62%	73.67%

	Model	Accuracy	Precision	Recall	F1 Score
3	Gradient Boosting	0.79	0.680000	0.566667	0.618182
0	Logistic Regression	0.78	0.666667	0.533333	0.592593
2	Random Forest	0.74	0.605263	0.383333	0.469388
1	Decision Tree	0.63	0.379310	0.366667	0.372881

Figure 5.1: Comparison of accuracy, precision, recall, and F1-score across classification models.

Gradient Boosting produced the strongest overall performance across the four reported metrics.

5.3 Final Model Selection

Gradient Boosting was selected as the final model.

Table 5.2: Final Model Performance

Model Property	Result
Selected Model	Gradient Boosting
Accuracy	79.00%
Precision	75.33%
Recall	72.62%
F1-score	73.67%
Validation Method	Five-Fold Stratified Cross- Validation
Tuning Method	GridSearchCV
Model Artifact model.pkl	final-gradient_boosting

5.4 Confusion Matrix and Error Analysis:

The final cost-sensitive threshold was set to 0.13, chosen because approving a defaulter (Type II error) costs 5 times more than rejecting a good applicant (Type I error), as specified in the project brief. This threshold was applied to the untouched 200-row test set to produce the confusion matrix below.

Table 5.3: Confusion Matrix at Threshold 0.13

Actual Outcome	Predicted Approve	Predicted Reject
Actual Good Applicant	63	77
Actual Default Applicant	5	55

Interpretation of the four cells:

- **63 (True Negative-equivalent):** Good applicants correctly approved
- **77 (Type I error):** Good applicants incorrectly rejected — a mild cost, since the bank only loses potential interest income
- **5 (Type II error):** Defaulters incorrectly approved — the costly mistake, since the bank loses the loan principal
- **55 (True Positive-equivalent):** Defaulters correctly rejected

Default Recall, the fraction of actual defaulters the model correctly caught, is calculated as:

$$\text{Default Recall} = \text{TP} / (\text{TP} + \text{FN}) = 55 / (55 + 5) = \mathbf{91.7\%}$$

The model identified 55 of the 60 applicants who actually defaulted, missing only 5. This high recall is a direct result of setting the threshold low (0.13 instead of the standard 0.50) — the model is deliberately tuned to be cautious, flagging anyone with even a modest chance of default, because missing a real defaulter is five times more expensive than an unnecessary rejection.

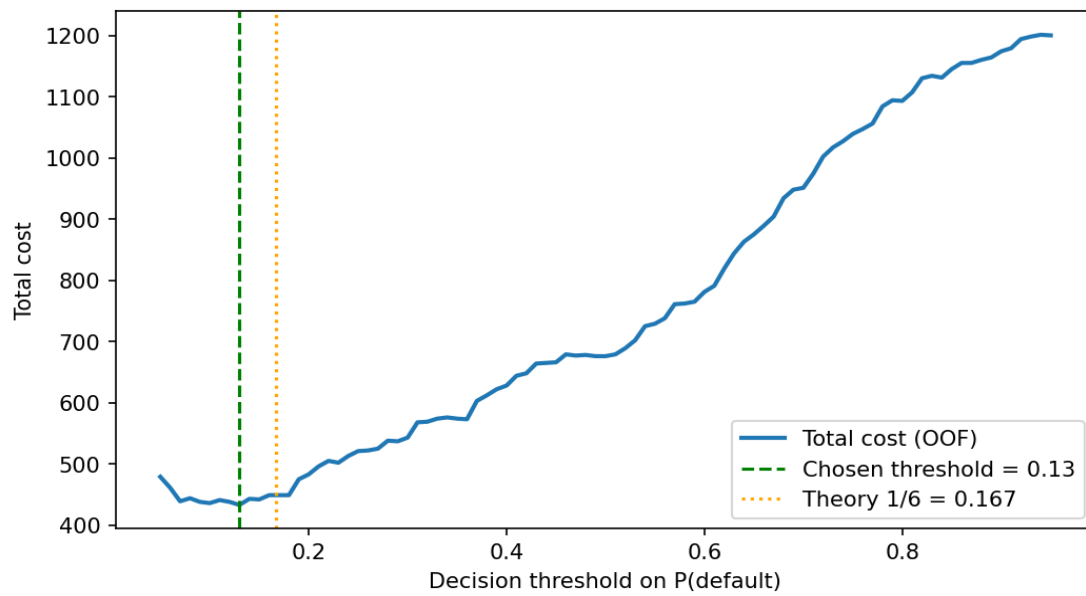
The trade-off: this caution comes at a cost. Of the 140 applicants who were actually good, 77 were still rejected — a Type I error rate that would be considered high under a normal (equal-cost) threshold, but is the expected and accepted consequence of prioritizing defaulter recall so heavily. Rather than treating every rejection as final, borderline applicants — particularly those in the Medium risk band ($0.15 \leq P(\text{default}) < 0.40$) — are routed to manual review instead of an automatic decision, so a human underwriter can make the final call on ambiguous cases instead of losing potentially good customers outright.

5.4 Cost Analysis

Table 5.4: Cost Comparison of Decision Strategies

Decision Strategy	Total Cost
Approve Every Applicant	300 units
Standard Threshold of 0.50	171 units
Cost-Tuned Threshold of 0.13	102 units

Figure 5.3: Expected decision cost at different probability thresholds.



The cost-sensitive threshold reduced total cost compared with both the approve-everyone strategy and the conventional 0.50 threshold.

CHAPTER 6

DASHBOARD, VISUALIZATION, AND IMPLEMENTATION

6.1 Dashboard and Visualization

The dashboard has three pages: Executive Overview, Risk Segmentation and a What-If Simulator. Executive Overview (Figure 6.1) shows four KPI cards, an expected loss comparison chart, and a Key Insights panel.



Figure 6.1: Executive Overview

Risk Segmentation (Figure 6.2) shows a cost-vs-threshold chart marking the chosen threshold, a confusion matrix, and applicant outcomes by risk band. A slicer filters all three visuals to one risk band at a time (Figures 6.3-6.5).

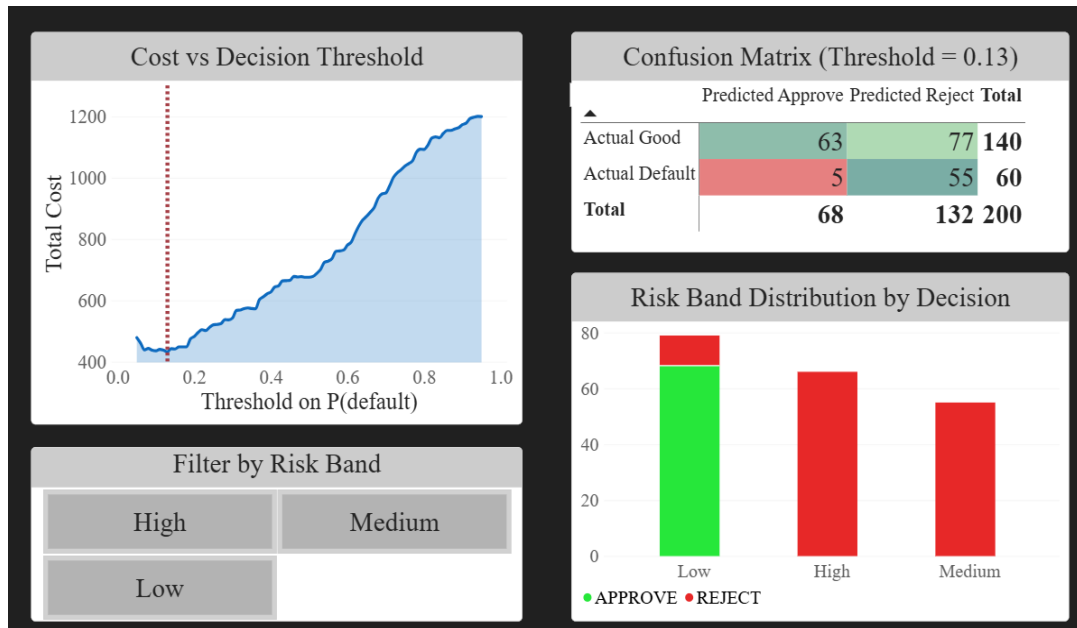


Figure 6.2: Risk Segmentation, unfiltered

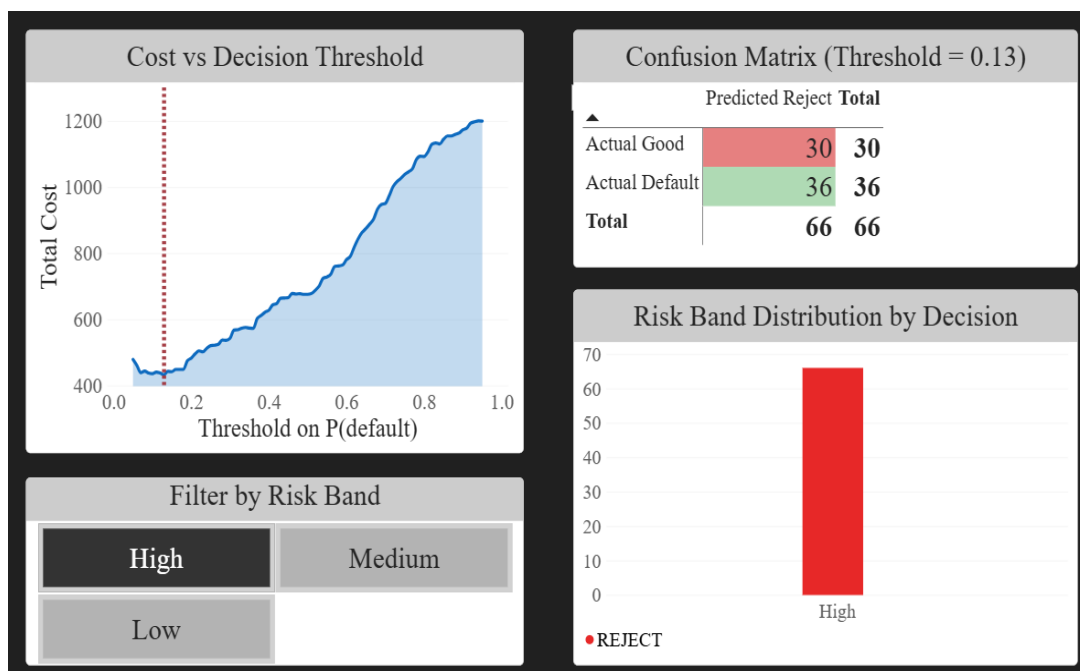


Figure 6.3: Filtered to High risk - all 66 rejected

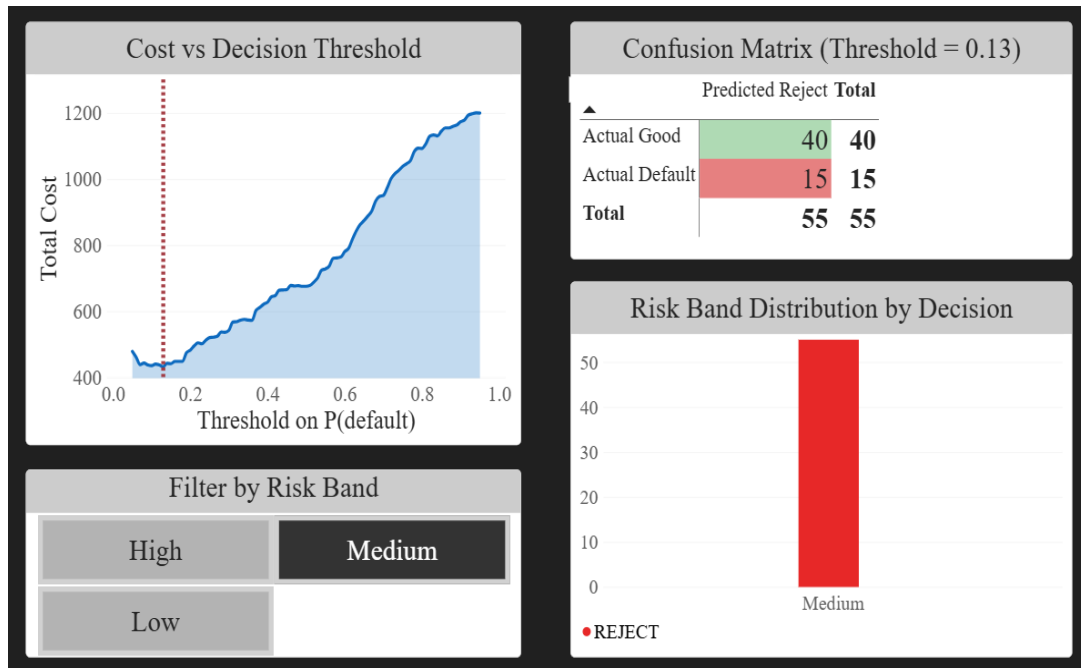


Figure 6.4: Filtered to Medium risk - all 55 rejected

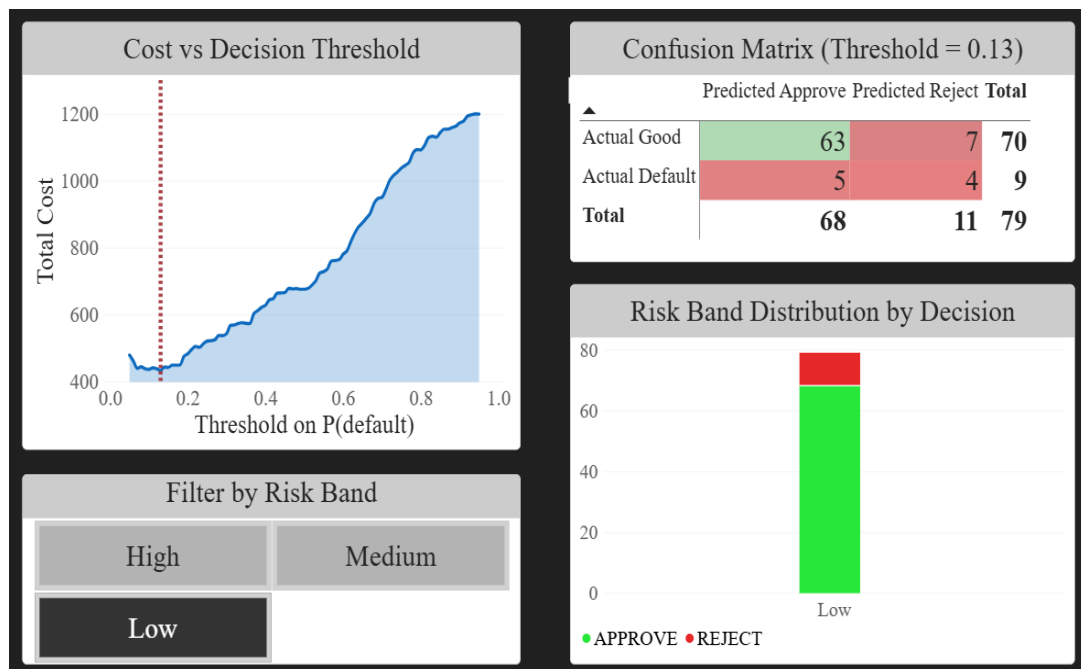


Figure 6.5: Filtered to Low risk - 68 approved, 11 rejected

The What-If Simulator lets the viewer drag a P(default) slider and see the resulting decision update live across all three tiers (Figures 6.6-6.8).

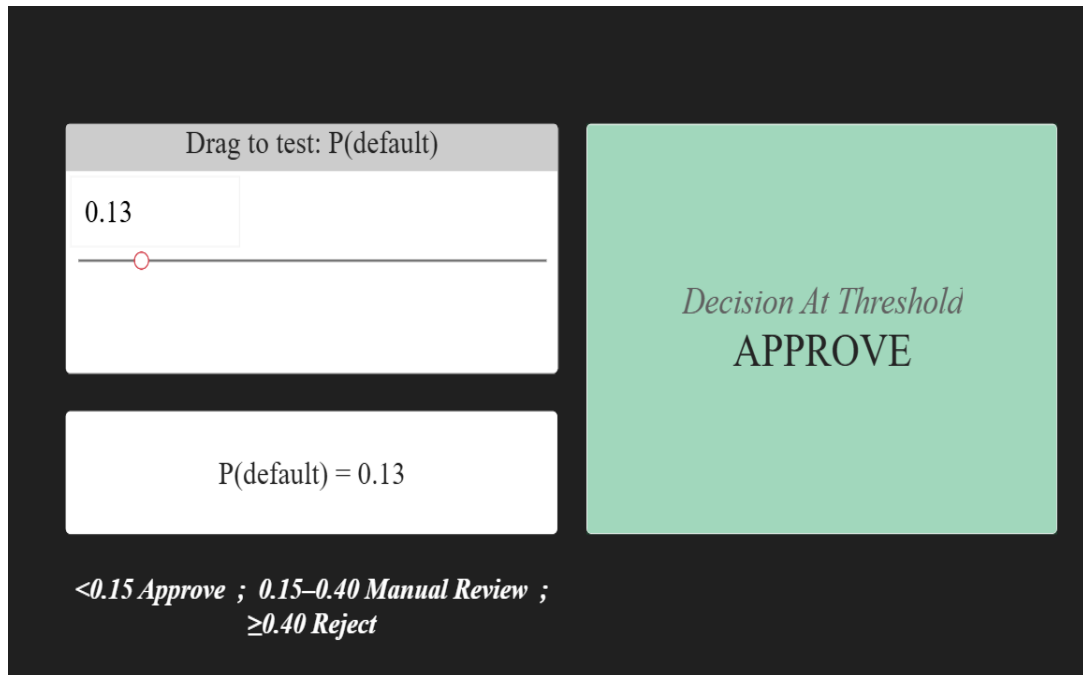


Figure 6.6: $P(\text{default}) = 0.13 \rightarrow \text{Approve}$

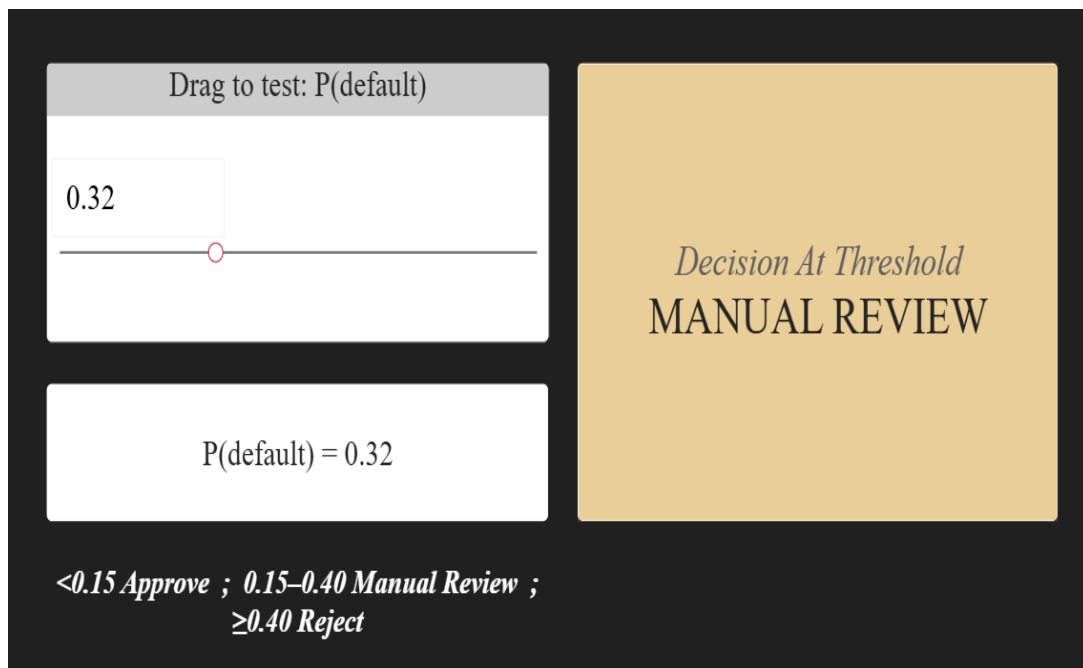


Figure 6.7: $P(\text{default}) = 0.32 \rightarrow \text{Manual Review}$

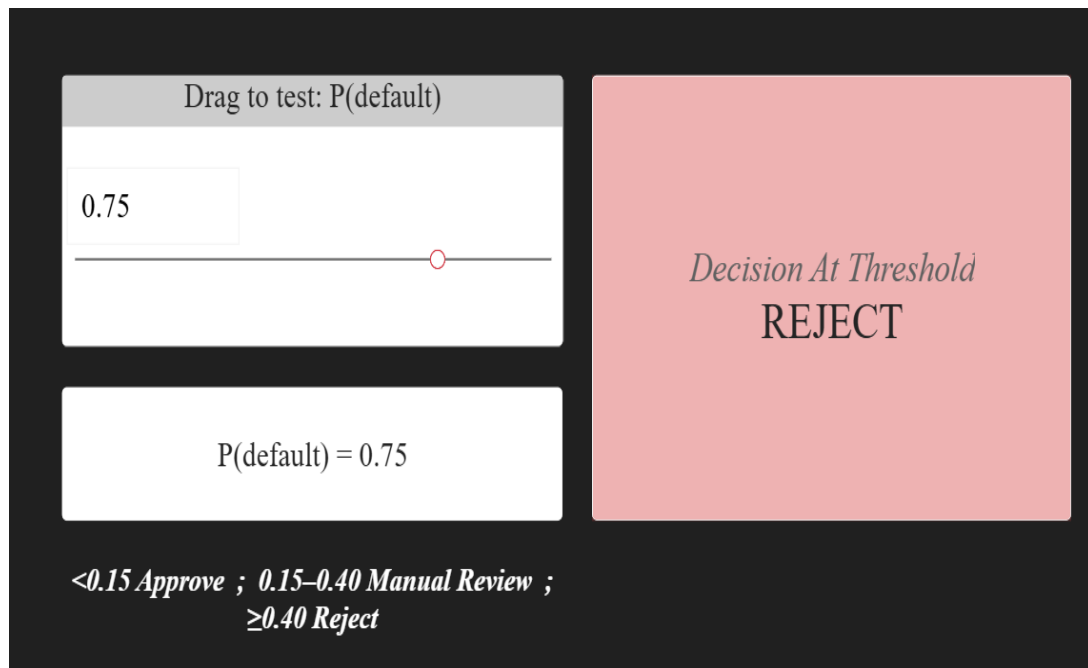


Figure 6.8: $P(\text{default}) = 0.75 \rightarrow \text{Reject}$

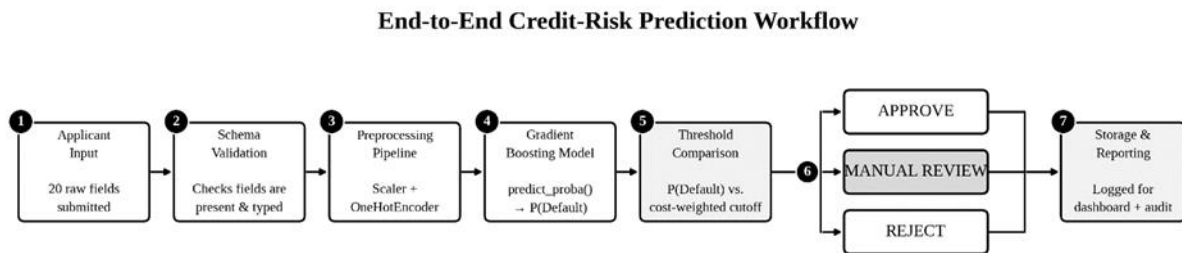
6.2 Important KPIs / Insights

- Approval Rate - 34% (68 of 200 applicants).
- Expected Loss (Approved Book) - Rs 7,194 vs Rs 1,67,231 if approving everyone (96% reduction).
- Defaulter Recall - 91.7% (55 of 60 defaulters caught).
- Cost Saved - 198 units (66%) vs approving everyone.

Insights

1. The tuned threshold (0.13) is close to the theoretical optimum (0.167), suggesting the model's probability estimates are well-calibrated.
2. The model rejects 66% of applicants, including 77 good ones - the trade-off of weighting a missed defaulter 5x costlier than a wrongly rejected good applicant.
3. Compared to the naive 0.50 threshold, tuning to 0.13 saves an additional 69 cost units (40%), showing the value of empirical threshold tuning over a default cutoff.

6.3 Prediction Workflow:



The complete workflow is:

Input → Preprocessing → Model → Prediction → Decision Output The

workflow consists of the following steps:

1. The user enters the applicant details.
2. The system validates the input schema.
3. The preprocessing pipeline transforms categorical and numerical values.
4. The Gradient Boosting model predicts the default probability.
5. The probability is compared with the selected threshold.
6. The system returns Approve, Manual Review, or Reject.
7. The output is stored for reporting and dashboard analysis.

6.4 Sample Input and Output

Table 6.2: Sample Prediction Results

Credit Amount	Default Probability	Predicted Class	Final Decision
1,200	0.016	Good	Approve
8,500	0.832	Bad	Reject
3,200	0.077	Good	Approve
9,200	0.249	Good	Manual Review
6,100	0.641	Bad	Reject

Sample Prediction Output

Predicted_Class	Probability_Class_1	Probability_Class_2	Prediction_Confidence
1	0.984427	0.015573	0.984427
2	0.168370	0.831630	0.831630
1	0.923147	0.076853	0.923147
1	0.751106	0.248894	0.751106
1	0.857885	0.142115	0.857885
2	0.359173	0.640827	0.640827
1	0.553782	0.446218	0.553782
1	0.823733	0.176267	0.823733
1	0.742590	0.257410	0.742590
2	0.359173	0.640827	0.640827

Figure 6.3: Sample output generated by the credit-risk prediction workflow.

CHAPTER 7

CHALLENGES, LIMITATIONS, AND IMPROVEMENTS

7.1 Challenges Faced

- The dataset contained both categorical and numerical variables.
- Data leakage had to be prevented during preprocessing and validation.
- The target variable was moderately imbalanced.
- Technical performance had to be balanced with financial consequences.
- Consistent feature order had to be maintained during inference.
- Multiple team members worked on a shared project repository.
- The original dataset did not contain direct financial-loss data.

7.2 Approaches That Did Not Work

The Decision Tree model performed less effectively than the other models.

Table 7.1: Performance of Selected Approaches

Model	Accuracy	Observation
Decision Tree	63%	Lower performance and possible overfitting
Gradient Boosting	79%	Better generalization and balanced metrics

A strict binary Approve or Reject system was also less suitable for borderline applicants. Consequently, a three-tier decision structure was recommended.

7.3 Current Limitations

- The training dataset is small.
- The data may not represent current Indian banking customers.

- Loss Given Default was assumed rather than learned from actual records.
- The 5:1 cost ratio is an academic assumption.
- The project does not include full production deployment.
- The project does not include complete fairness and regulatory analysis.
- The dashboard uses scored data rather than a live data stream.

7.4 Future Improvements

- Use larger and more recent banking datasets.
- Integrate credit-bureau scores and transaction history.
- Use actual bank-specific recovery and loss values.
- Perform fairness and bias testing.
- Apply SHAP for explainable predictions.
- Deploy the model using Flask or FastAPI.
- Containerize the application using Docker.
- Add data-drift and prediction-drift monitoring.
- Retrain the model periodically using new repayment outcome

CHAPTER 8

TEAM CONTRIBUTION AND LEARNING

8.1 Team Roles and Contributions

Table 8.1: Team Roles and Contributions

Student Name	Register Number	Assigned Role	Major Contribution	Tools and Skills Used
Batchu Lekhya Reddy	241BCADA41	Data Engineer	Validated the dataset, checked missing values and duplicates, performed schema validation, and applied preprocessing.	Python, Pandas, NumPy, Scikit-learn
Syed Farhan Basith Madni	241BCADA49	Data Analyst	Conducted exploratory data analysis, skewness analysis, outlier analysis, and correlation analysis.	Pandas, Matplotlib, Seaborn, Statistics
Patricia F T	241BCADA27	Data Scientist	Trained and compared models and performed cross-validation and hyperparameter tuning.	Scikit-learn, GridSearchCV, Model Evaluation
Rajeshwari S S	241BCADA01	Machine Learning Engineer	Developed the inference workflow and validated model input and output.	Python, Joblib, Logging
Roopashree P	241BCADA38	Analytics Engineer	Developed cost-sensitive threshold analysis and business decision rules.	Python, Excel, Cost Analysis
Gedela Nikhita	241BCADA65	BI Developer	Designed the Power BI dashboard and created business KPIs.	Power BI, DAX, Data Visualization

8.2 Skills and Learning Outcomes

The project provided practical learning in:

- Data cleaning and validation.
- Categorical and numerical preprocessing.
- Exploratory data analysis.
- Machine learning model comparison.
- Cross-validation and hyperparameter tuning.
- Classification evaluation.
- Cost-sensitive decision analysis.
- Inference pipeline development.
- Dashboard design using Power BI.
- Collaborative project development using GitHub.

CONCLUSION

The project addressed the problem of retail banking credit-risk prediction using the UCI Statlog German Credit dataset. The objective was to identify applicants who may represent a high risk of default and convert model predictions into practical lending decisions.

The project followed a complete machine learning methodology involving data cleaning, prepro-cessing, exploratory analysis, feature transformation, model comparison, hyperparameter tuning, threshold optimization, inference development, and dashboard reporting.

Four classification algorithms were evaluated: Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting. Gradient Boosting was selected as the final model because it achieved the strongest overall performance, with 79% accuracy, 75.33% precision, 72.62% recall, and 73.67% F1-score.

The most important result was the use of a cost-sensitive threshold of 0.13. This threshold improved the identification of actual defaulters and reduced expected cost compared with both the standard threshold and the approve-everyone baseline. The model identified 55 out of 60 defaulters, giving a default recall of 91.7%.

The project demonstrates that a credit-risk model should not be evaluated only on accuracy. In financial applications, the consequences of incorrect decisions must also be incorporated into the decision process.

- [1] H. Hofmann, “Statlog German Credit Data,” UCI Machine Learning Repository, 1994.
Available at:
<https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data>
- [2] F. Pedregosa *et al.*, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [3] W. McKinney, “Data Structures for Statistical Computing in Python,” in *Proceedings of the 9th Python in Science Conference*, 2010, pp. 56–61.
- [4] Scikit-learn Developers, “Scikit-learn Documentation.” Available at: <https://scikit-learn.org/>
- [5] Pandas Development Team, “Pandas Documentation.” Available at:
<https://pandas.pydata.org/>
- [6] Microsoft, “Power BI Documentation.” Available at:
<https://learn.microsoft.com/power-bi/>
- [7] St. Joseph’s University, Department of Advanced Computing, “BCADA 5P126 Machine Learning Lab Project Guide,” 2026.

GITHUB REPOSITORY

The project repository is available at:

<https://github.com/SSRajeshwari-13/ML-Final-Lab-Group-09>

The screenshot shows a GitHub repository interface. At the top, there is a list of files and folders with their descriptions and last update times:

File/Folder	Description	Last Update
dashboard	Power BI visualizations	10 minutes ago
data	All the processed datasets are available here	9 minutes ago
notebooks	The entire Machine Learning project pipeline notebook	8 minutes ago
presentation	Client pitch presentation	5 minutes ago
report	Project report	3 minutes ago
src	All the required source codes are uploaded here	2 minutes ago
tests	Add sanity tests for model and preprocessor	last week
weekly_report	Weekly report updates	1 minute ago
README.md	Update README with project details and structure	yesterday
requirements.txt	Add pinned Python dependencies for MLE inference	last week

Below this list, the 'README' file is selected and expanded. The README content includes:

ML-Final-Lab-Group-09

CrediTrust Capital — AI-Powered Credit Risk Scoring

CrediTrust Capital — AI-powered credit risk scoring for retail lending, turning applicant data into fast, defensible Good/Bad credit-risk decisions.

Figure A.1: Screenshot of the GitHub repository containing the project source code and documentation

APPENDIX B IMPORTANT CODE

SNIPPETS

C.1 Cost-Sensitive Threshold Logic

```
1 cost_fp = 1
2 cost_fn = 5
3
4 threshold = cost_fp / (cost_fp + cost_fn)
5
6 if probability_default > threshold:
7     decision = "Reject"
8 else:
9     decision = "Approve"
```

C.2 Three-Tier Decision Logic

```
1 if probability_default < 0.13:
2     decision = "Approve"
3 elif probability_default < 0.25:
4     decision = "Manual Review"
5 else:
6     decision = "Reject"
```

C.3 Prediction Workflow

```
1 input_data = validate_schema(input_data)
2
3 processed_data = preprocessor.transform(input_data)
4
5 probability_default = model.predict_proba(
6     processed_data
7 )[:, 1]
8
9 predicted_class = model.predict(processed_data)
```